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Draft NPS/002/024 Technical Specification for Fibre Optic Cables

1. Purpose

The purpose of this document is to detail the technical requirements for fibre optic cables used to provide communication and protection links on the CE Electric UK distribution system.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title

2. Scope

This specification is split into two parts

- a) The specification for Optical Fibres
- b) The specification for Fibre Optic cables (Host cable to carry the fibres)

The Fibre optic cables detailed in this specification are for installation by direct burial, laid directly in closed cable trenches or installation into cable duct systems. These cables will generally be used to provide unit protection between primary substation or switching points or in substation compounds where high induced voltages or rise of earth potential could present a problem if the cables contained traditional metallic components.

This cable is also suitable for applications other than telecommunications and protection where equipment requires optical connections on a substation site.

3. Technical Specification

3.1 Optical Fibres

3.1.1 Informative: The mechanical construction of optical cables used by CE Electric UK has an optical core with loose buffer tubes containing the optical fibres. The construction of the cable can be the same design and specification whatever fibre type it contains within the buffer tubes. There are 3 types of fibre used in the construction of cables; multimode fibre, Non Dispersion Shifted Fibre NDSF single-mode fibre, and Non Zero Dispersion Shifted Fibre NZDSF single mode fibre.

Multimode fibres are used for short distance applications typically less than 2 km or within a substation for communication between items of equipment. In most applications 62.5/125 cm fibre multi-mode will be specified as this provides a better better optical solution due to the availability of cheaper light source terminating equipment.

Single mode NDSF fibre is used for most applications operating at STM16 (2.5 Gbps) or less. Systems

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operating at bit rates above STM16 will normally specify NZDSF fibre.

By default the Optical Fibres shall be Single Mode fibres to G652D or G655 and BSEN 60793-2-50:2008 type B1.3 "Product Specification – Sectional Specification for class B single mode Fibres"
Unless specified otherwise within the project specification, the cable shall contain a minimum of 18 fibres arranged in 3 tubes with all fibres and tube colours compliant with EIA-598be (See Appendix 2 for details)

Where fibres circuits are being considered for use within a substation site or on projects with distances of less than 2km the fibre shall be specified as multimode 62.5 /125µm glass fibre Type A1b to BSEN 60793-2-10 2007 and BSEN 60794-3-10 2002. Additionally all fibres shall conform to ITU-T G651 and be suitable for a ducted installation.

- 3.1.2 The optical fibre coating material shall be mechanically strippable.
- 3.1.3 The optical fibres shall be capable of being joined by the fusion splicing technique. The splices shall be capable of meeting the performance requirements detailed in the CE Electric UK specification.
- 3.1.4 The grade of fibre required will be stated in the supply contract. The Manufacturer should be prepared to offer multimode fibre suitable for use at 850 and 1310 nm, conventional single-mode NDSF fibre for use at 1310 and 1550 nm, and NZDSF fibre for long distance transmission at 1550 nm.
- 3.1.5 Details of the optical fibre including the optical fibre type, the attenuation (dB/km), the refractive index of the optical core, the fibre manufacturer and the manufacturing process shall be supplied in accordance with Appendix 2.
- 3.1.6 Where a buffer tube contains multiple fibres the fibres shall be coloured in accordance with – (EIA-598be) Fibre Optic Colour Code Chart sequence:

COLOR CODING SCHEME

When multiple fibres are housed in a single buffer tube, the colour coding of the fibre optic cables enables the installer/administrator to easily identify the individual fibres. This colour scheme may also be used, but is not mandatory for fibres that are identified by their position; eg: ribbon fibre.

When a number of fibres (e.g., 6, 12, 18, 24, 36) form a unit/bundle, and units/bundles are grouped to form groups, the units and groups shall be uniquely identified and shall also follow the colour scheme shown in the table below.

Fibre Colour
1 Blue
2 Orange
3 Green
4 Brown
5 Grey
6 Yellow
7 Red
8 Violet
9 Pink
10 White
11 Black
12 Turquoise

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3.2 Fibre Optic Cables

Shall be non armoured cables manufactured to BSEN 60794-3-10 "Outdoor cables-Family specification for duct and directly buried optical telecommunication cables" and in accordance with the following technical requirements

- 3.2.1 The cable shall be circular in cross section and free from pinholes, joints, repairs and other defects. Materials used in the construction of the cable shall not affect the physical or optical properties of the fibres and shall be compatible with each other.
- 3.2.2 The required number of fibres will be specified in the individual contract document. Where this has not been specified, the default cable design shall incorporate an 18 fibre cable. The cable shall be arranged with a minimum of 3 tubes each containing 6 fibres per tube.
- 3.2.3 The tubes shall be colour coded for ready identification at either end. Where the tubes are arranged in a circular configuration it is acceptable to mark the tubes to indicate the start and stop of the circle.
- 3.2.4 The preferred material for the outer sheath of the cable is High Density Polyethylene with a 2.5% loading of carbon black (well dispersed) for weather resistance purposes. The composition of the polyethylene sheath shall comply with Section 4 of BS 6234 and conform to the physical and mechanical requirements detailed in Appendix 2 of this specification.
- 3.2.5 Other sheath materials offering similar properties to polyethylene shall only be used with the approval of CE Electric UK.
- 3.2.6 The outer sheath shall be clearly marked in white with the following information every metre

llll M CE Electric UK Optical Cable *ffff* F *tttt* logo *yyyy*

llll - Sequential Length Mark
ffff - Fibre Count
tttt - Fibre Type e.g. 9/125
logo - Manufactures Logo
yyyy - Year of Manufacture

- 3.2.7 A tape shall be laid under the inner sheath to identify the manufacturer and the year of manufacture.
- 3.2.8 Before shipment, the cable ends shall be capped to prevent the ingress of water and marked or coloured to distinguish between the directions of rotation of the colour code as described in BS 3573.

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3.3 Performance Requirements

- 3.3.1 The cable shall conform to the test requirements detailed in clause 3.4 and the performance parameters in detailed in Appendix 2.
- 3.3.2 The optical fibres shall not be subjected to any critical stress when the cable is wrapped and unwrapped around a mandrel of twelve times the diameter of the cable for four complete turns.
- 3.3.3 The cable shall withstand a crush load of 1000 N load applied at right angles laterally, via a 25mm diameter rod, without damage to the optical fibres. Alternatively the cable shall withstand a crush load of 5000 N applied with a 100mm flat plate.
- 3.3.4 Informative: it is preferred that the crush test is performed using a 25mm round bar to simulate a load from a second cable crossing at right angles under external pressure. A load of 5000 N applied with a 100mm flat plate has found to be of similar severity.
- 3.3.5 The fibre tubes shall be filled with a water-blocking compound to prevent water penetration and for shock absorption. The filling compound shall pass the drip test as specified in section 6.
- 3.3.6 A water blocking compound shall be used in the interstices of the cable or alternatively a water blocking tape may be used beneath the outer sheath.
- 3.3.7 The cable shall be designed such that cable strain is not directly imparted to the optical fibres during normal service and the maximum transient strain imparted on the fibre during installation will not exceed 0.2"
- 3.3.8 The optical fibre cables shall have an installed design life of at least 25 years

3.4 Test Requirements

Type Approval Tests

- 3.4.1 The tenderer shall provide evidence to establish the suitability of the products being offered for the purpose. Where the product has not already been recorded on the CE Electric UK assessed product database, then the tenderer shall provide copies of type tests to confirm compliance with this specification.

3.4.2 Routine Tests

All cables shall be subjected to inspection and test to ensure that the type registered standard is being maintained. The Manufacturer shall propose a series of tests to prove that the design performance is being achieved. For the optical fibre cable a performance test will normally be required at the factory, after delivery and after installation to ensure that no degradation has occurred between manufacture and commissioning.

3.4.3 Site Tests

The installed optical cable shall be tested with an OTDR on each fibre core from each end after installation, to characterise the attenuation of the installation and to ensure that no physical damage has occurred to the fibre during installation.

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An end to end attenuation measurement shall be taken in each direction using an optical source and an optical power meter. The overall attenuation of the installed optical cable shall not exceed that calculated using the attenuation and splice loss values specified in the project specification.

4. References

4.1 External Documentation

Reference	Title
BS 6234	Specification for polyethylene insulation and sheath of electric cables
BS 3573	Specification for polyolefin copper-conductor telecommunication cables
BSEN 60794-3-10	Outdoor cables-Family specification for duct and directly buried optical telecommunication cables
BSEN 60793-2-10 2007	Optical fibres. Product specifications. Sectional specification for category A1 multimode fibres
BSEN 60793-2-50:2008	Optical fibres. Product specifications. Sectional specification for class B single-mode fibres
G652D	This single-mode optical fibre has significantly reduced optical attenuation at water absorption wavelength around 1383nm. It provides expanded transmission window from 1310nm to 1550nm continuously. This fibre has low attenuation over a wide wavelength range and its high performance coating ensures long term reliability.
EIA-598be	Fibre Optic Colour Code Chart

4.2 Internal documentation

Reference	Title
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5. Definitions

Term	Definition
OTDR	Optical time-domain Reflectometer - An OTDR tester is used for testing and fault finding within fibre optic networks and fibre optic cables.
Interstices	Gap between strands or tubes in a round cable

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Line Standards Manager		

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M Nicholson	Head of System Strategy		

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
J Paine	Technical Services Manager		

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator		

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards & Assessment Manager		

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement		

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Record of Revision

This issue has been revised and amended as follows:

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Appendix 1 – Technical Data

Optical Fibre Performance requirements

Multimode cable shall comply with the following requirements

Parameter	Limit	Specification		Test Method	
		IEC 60793-2	ITU-T G651	IEC 60793-1	ITU-T G651
Coating diameter	245 ± 10 µm	4.1.4	-	A2	6.4
Cladding diameter	125 ± 2 µm	4.1.1.4	3.1.2	A2	6.4
Cladding non-circularity	≤ 2 %	4.1.1.4	3.1.4.2	A2	6.4
Core non-circularity	≤ 6%	4.1.1.4	3.1.4.1	A2	6.4
Core/cladding concentricity error	≤ 6%	4.1.1.4	3.1.3	A2	6.4
Proof Strain	≥ 1%	4.1.3	3.3.3	B1	G650 2.6
Stripping force	≤ 3.2 N				
Chromatic dispersion 850nm	≤120 ps/(km.nm)	4.1.4	4.2.3	C5A	6.9
Chromatic dispersion 1300 nm	≤ 3.5 ps/(km.nm)	4.1.4	4.2.3	C5A	6.9

In addition 50/125 µm fibres:

Core diameter:	50 ± 3 µm	4.1.1.4	3.1.1	A2	6.4
Attenuation 850 nm	≤ 2.5 dB/km	4.1.4	4.1	C1A, C1C	6.5, 6.6, 6.8
Attenuation 1300 nm	≤ 0.8 dB/km	4.1.4	4.1	C1A, C1C	6.5, 6.6, 6.8
Bandwidth 850 nm	≥ 250 MHz.km	4.1.4	4.1.4	C2B	6.9
Bandwidth 1300 nm	≥ 1000 MHz.km	4.1.4	4.1.4	C2B	6.9
Numerical Aperture	0.22 ± 0.02	4.1.4	4.1.4	C6	6.4

In addition 62.5/125 cm fibres:

Core diameter:	62.5 ± 3 µm	4.1.1.4	NA	A2	6.4
Attenuation 850 nm	≤ 3.0 dB/km	4.1.4	4.1	C1A, C1C	6.5, 6.6, 6.8
Attenuation 1300 nm	≤ 1.0 dB/km	4.1.4	4.1	C1A, C1C	6.5, 6.6, 6.8
Bandwidth 850 nm	≥ 200 MHz.km	4.1.4	NA	C2B	6.9
Bandwidth 1300 nm	≥ 600 MHz.km	4.1.4	NA	C2B	6.9
Numerical Aperture	0.275 ± 0.015	4.1.4	NA	C6	6.4

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The following requirements also apply to single-mode NDSF optical fibres:

Parameter	Limit	Specification IEC 60793-2	ITU-T G652	Test Method IEC 60793-1	ITU-T G650
Coating Diameter	245 ± 10 µm	5.1.4	-	A3	-
Cladding Diameter	125 ± 2 µm	5.1.4	5.2	A2	2.2.1
Cladding Non-circularity	≤ 2 %	5.1.4	5.4.2	A2	2.2.1
Mode field Diameter	9 ± 0.5 µm	5.1.4	5.1	C9A	2.1.1
Mode Field Concentricity.	≤ 1µm	5.4.3	5.3	A2	2.2.1
Proof Strain	≤ 1 %	5.3	5.7.3	B1	2.6
Stripping force	≤ 3.2 N	-	-	-	-
Attenuation 1300 nm	≤ 0.4 dB/km	5.4.1	6.1	C1A	2.4.1
Attenuation 1550 nm	≤ 0.25 dB/km	5.4.1	6.1	C1A	2.4.1
Chromatic dispersion 1285 –1330 nm	≤ 3.5 ps/(km.nm)	5.4.2	6.2	C5C	2.5.1
Chromatic dispersion 1550 nm	≤ 20 ps/(km.nm)	5.4.2	6.2	C5C	2.5.1
Zero dispersion wavelength	1310 nm	5.4.2	6.2	C5B	-
Cut-off wavelength	≤1270 nm	5.4.4	5.5	C7B	2.3.1
Polarisation mode distortion individual fibre length	≤ 0.5ps/_km	-	6.3	IEC TS61941	2.7
Polarisation mode distortion link of concatenated fibres	≤ 0.1ps/_km	-	6.3	IEC TS61941	2.7

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The following requirements also apply to single mode NZDSF optical fibres

Parameter	Limit	Specification IEC 60793-2	ITU-T G652	Test Method IEC 60793-1	ITU-T G650
Coating Diameter	245 ± 10 µm	5.1.4	-	A3	-
Cladding Diameter	125 ± 2 µm	5.1.4	5.2	A2	2.2.1
Cladding Non-circularity	≤ 2 %	5.1.4	5.4.2	A2	2.2.1
Nominal Mode field Diameter	8 to 11µm	5.1.4	5.1	C9A	2.1.1
Mode Field Concentricity.	≤ 1µm	5.4.3	5.3	A2	2.2.1
Proof Strain	≤ 1 %	5.3	5.7.3	B1	2.6
Stripping force	≤ 3.2 N	-	-	-	-
Attenuation 1525 - 1575nm	≤ 0.25 ± 0.05 dB/km	5.4.1	6.1	C1A	2.4.1
Chromatic dispersion 1525 - 1575 nm	≤ 1-6 ps/(km.nm)	5.4.2	6.2	C5C	2.5.1
Cut-off wavelength	≤1270 nm	5.4.4	5.5	C7B	2.3.1
Polarisation mode distortion individual fibre length	≤ 0.25ps/√km	-	6.3	IEC TS61941	2.7
Polarisation mode distortion link of concatenated fibres	≤ 0.8ps/√km	-	6.3	IEC TS61941	2.7

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Appendix 2 – Technical Data

Optical Fibre Cable

The following design constraints apply and approval tests shall prove that the optical fibre cable meets these requirements:

Maximum outer diameter 20 mm

Maximum weight (W) 400 kg/km

Conformity of sheath thickness ± 1 mm

Operating temperature range - 25 to 40°C

Cable Parameter	Limit	IEC 60794-1
Tensile performance	$\geq 2 \times W \times 9.81 \text{ N}$	E1
Abrasion resistance sheath	≥ 300 cycles	E2A
Abrasion resistance marking	≥ 300 cycles	E2B
Crush 25 mm round bar	$\geq 1000 \text{ N}$	E3
Crush 100 mm flat plate	$\geq 5000 \text{ N}$	E3
Impact	$\geq 50 \text{ Nm}$	E4
Torsion $\pm 90^\circ$	10 cycles	E7
Cable bend radius 12 x dia	4 turns 10 cycles	E11
Cut through resistance	$\geq 500 \text{ N/mm}$	E12
Tear resistance	$\geq 30 \text{ N/mm}$	BS6469
Temperature cycling; 4 cycles	- 25 +40°C	F1
Water penetration	–	F5
Bleeding of tube compound 24 h 60°C	none	E12